

# DSA0601 Data Handling and Visualization

## Slot A

### CO2 – Assessment Test 1 (AT1)

#### Chart Construction Test

#### Instructions:

For each question, write the program in the specified language (R or Python), generate the required visualization, and answer the interpretation sub-questions clearly with justification. DSA06 – Data Handling and Visualization | Assessment Tool: Experiment

## DSA06 – DATA HANDLING AND VISUALIZATION

### CO3\_AT1\_Assessment Tool: EXPERIMENT

**Topics Covered:** ECDF and Q-Q Plots; Multiple Distributions; Vertical-Axis Distributions; Bean Plots; Proportions; Pie/Side-by Side/Stacked Bars; Stacked Densities; Treemaps; Sunburst Charts; Parallel Sets; Sankey Diagrams; Nested Proportions; Avoiding Misleading Proportions.

**General Instruction:** For each experiment, use an appropriate visualization, justify the design, interpret the result, and identify potential visualization bias. R/ggplot2 or an equivalent visualization tool may be used.

### Question 1 – Distribution Diagnosis using ECDF and Q-Q Plot

**Scenario:** An engineering department records response time (ms) of an automated fault-detection system under 20 test runs. The team claims the distribution is approximately normal and wants to use mean and standard deviation for monitoring.

**Dataset (n=20):** 118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162, 171, 184,

205, 249, 318 **Tasks:**

1. Construct an ECDF and a normal Q-Q plot.

#### ECDF and Normal Q-Q Plot

#### Dataset:

118, 121, 125, 127, 129, 131, 134, 136, 138, 141, 144, 147, 151, 156, 162, 171, 184, 205, 249, 318 For the

**ECDF**, the x-axis represents response time and the y-axis represents the cumulative proportion of observations. For the **normal Q-Q plot**, the observed response times are compared with theoretical normal-distribution quantiles.

2. Use both plots to evaluate the normality claim.

#### Evaluation of the Normality Claim

The data do not appear to follow a normal distribution.

- The ECDF shows that most observations are concentrated between approximately 118 and 184 ms, followed by a long upper tail.
- The Q-Q plot would show noticeable departures from the straight reference line, particularly at the upper end.
- Therefore, the assumption of approximate normality is questionable.

3. Identify evidence of skewness and/or extreme observations.

## Skewness and Extreme Observations

The distribution shows **strong positive (right) skewness**.

Important observations in the upper tail are:

- **205 ms**
- **249 ms**
- **318 ms**

The sample statistics are approximately:

| Statistic          | Value     |
|--------------------|-----------|
| Mean               | 159.35 ms |
| Median             | 142.5 ms  |
| Standard deviation | 49.00 ms  |
| Skewness           | 2.25      |

Since the **mean > median** and skewness is strongly positive, there is clear evidence of right skewness. Using the IQR rule:

- $Q1 = 130.5$  ms
- $Q3 = 164.25$  ms
- $IQR = 33.75$  ms
- Upper fence = 214.875 ms

Thus, **249 ms and 318 ms** can be considered extreme observations by the IQR criterion.

4. Recommend percentile-based limits or mean  $\pm$  standard-deviation limits, with justification.

**Key competency:** ECDF, Q-Q plot, skewness, outliers, distribution interpretation

## Recommended Monitoring Limits

**Percentile-based limits are recommended** rather than mean  $\pm$  standard deviation.

The reason is that the response-time distribution is strongly right-skewed and contains extreme high values. Mean

$\pm$  SD limits are sensitive to these observations and rely on a roughly symmetric/normal distribution.

For example, empirical percentile limits could be used:

- **90th percentile  $\approx$  209.4 ms**
- **95th percentile  $\approx$  252.45 ms**

So, a practical monitoring approach could use a **95th-percentile upper limit of about 252 ms**, depending on the department's desired false-alarm rate.

## Conclusion

**The normality claim should be rejected or treated with caution.** The ECDF indicates a long right tail, while the Q-Q plot would show substantial deviation from normality at the upper end. The data have **positive skewness ( $\approx 2.25$ )** and extreme observations at **249 and 318 ms**. Therefore, **percentile-based monitoring limits are preferable to mean  $\pm$  standard deviation limits** for this dataset.

## Question 2 – Comparing Many Distributions and Bean Plots

**Scenario:** A semiconductor manufacturer measures chip fabrication cycle time (minutes) for four production lines. Management wants to identify differences in variability and distribution shape.

| Production Line | Cycle Times (minutes)                  |
|-----------------|----------------------------------------|
| L1              | 42, 44, 45, 46, 47, 48, 49, 50, 51, 53 |
| L2              | 38, 41, 44, 47, 50, 53, 56, 59, 62, 65 |
| L3              | 45, 46, 46, 47, 47, 48, 48, 49, 49, 50 |
| L4              | 40, 41, 42, 43, 44, 45, 48, 55, 61, 72 |

### Tasks:

1. Create a visualization comparing all four distributions.

### Visualization comparing all four distributions

A **box plot/violin plot (bean-plot equivalent)** is suitable because it allows the center, spread, and shape of all four production-line distributions to be compared on the same graph.

# Data

```
L1 <- c(42, 44, 45, 46, 47, 48, 49, 50, 51, 53)
```

```
L2 <- c(38, 41, 44, 47, 50, 53, 56, 59, 62, 65)
```

```
L3 <- c(45, 46, 46, 47, 47, 48, 48, 49, 49, 50)
```

```
L4 <- c(40, 41, 42, 43, 44, 45, 48, 55, 61, 72)
```

```
# Combine data
```

```
cycle_time <- c(L1, L2, L3, L4)
```

```
line <- factor(c(
```

```
  rep("L1", 10),
```

```
  rep("L2", 10),
```

```
  rep("L3", 10),
```

```
  rep("L4", 10)
```

```
))
```

```
data <- data.frame(line, cycle_time)
```

```
# Violin plot
```

```
boxplot(cycle_time ~ line,
```

```
  data = data,
```

```
  main = "Cycle Time Distribution Across Production
```

```
  Lines", xlab = "Production Line",
```

```
  ylab = "Cycle Time (minutes)",
```

```
  col = "lightblue")
```

```
# Add individual observations
```

```
stripchart(cycle_time ~ line,
```

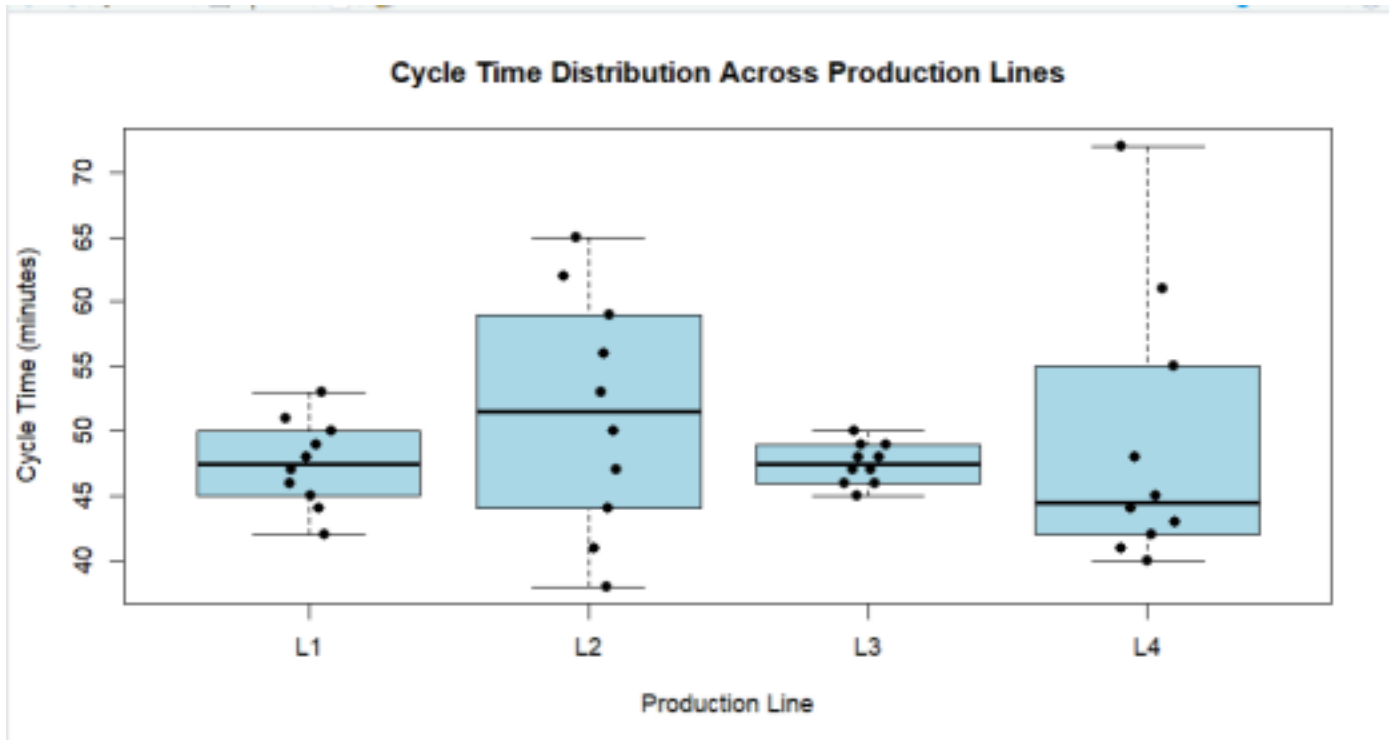
```
  data = data,
```

```
  vertical = TRUE,
```

```
  method = "jitter",
```

```
  pch = 19,
```

```
  add = TRUE)
```



### Expected interpretation

- **Greatest variability: L4**
  - Range =  $72 - 40 = 32$  minutes
  - SD  $\approx 10.44$  minutes
- **Most concentrated: L3**
  - Range =  $50 - 45 = 5$  minutes
  - SD  $\approx 1.58$  minutes
- **L1 and L3 have the same mean (47.5 min)**, but L3 is much more concentrated. This shows why **mean alone is insufficient** for comparing distributions.

**Conclusion:** L4 has the highest variability, while L3 has the most concentrated cycle-time distribution.

2. Construct bean plots or an equivalent distribution visualization along the vertical axis

Distribution comparison

| Line | Mean (min) | Std. Dev. (min) | Range |
|------|------------|-----------------|-------|
| L1   | 47.5       | 3.37            | 11    |
| L2   | 51.5       | 9.08            | 27    |
| L3   | 47.5       | 1.58            | 5     |
| L4   | 49.1       | 10.44           | 32    |

3. Determine which line has greatest variability and which has the most concentrated distribution.

### Greatest variability and most concentrated distribution

- **Greatest variability: L4**
  - Standard deviation = **10.44 min**

- Range = **32 min**
- The high values **55, 61, and 72** create a long upper tail.
- **Most concentrated: L3**
  - Standard deviation = **1.58 min**
  - Range = **5 min**
  - Most observations lie tightly between **45 and 50 minutes**.
- **L2** also has substantial variability and is spread fairly evenly from 38 to 65 minutes.
- **L1** has relatively low variability and is fairly compact around 47–50 minutes.

4. Explain why comparing only means is insufficient.

**Key competency:** Multiple distributions, vertical-axis distributions, bean plots, variability

## Why comparing only means is insufficient

Comparing only means can hide important differences in the distributions.

For example:

- **L1 mean = 47.5 min**
- **L3 mean = 47.5 min**

They have **exactly the same mean**, but their variability is very different:

- **L1 SD = 3.37 min**
- **L3 SD = 1.58 min**

Similarly, L4 has a mean of only **49.1 min**, but its distribution is much more variable because of the high values

up to **72 min**. **Conclusion**

**L4 has the greatest variability**, while **L3 has the most concentrated distribution**. A bean plot or equivalent distribution visualization is better than comparing means alone because it reveals **spread, concentration, skewness, and extreme observations**.

## Question 3 – Proportion Visualization and Choice of Chart

**Scenario:** A university analyzes preferred learning mode of 240 students across three departments. The dean wants comparison without exaggerating small differences.

DSA06 – Data Handling and Visualization | Assessment Tool: Experiment

| Department | Online (%) | Blended (%) | Face-to-Face (%) |
|------------|------------|-------------|------------------|
| CSE        | 52         | 28          | 20               |

|     |    |    |    |
|-----|----|----|----|
| ECE | 45 | 35 | 20 |
|-----|----|----|----|

|    |    |    |    |
|----|----|----|----|
| IT | 30 | 40 | 30 |
|----|----|----|----|

### Tasks:

1. Create a side-by-side bar chart and a 100% stacked bar chart.

# Data

```
dept <- c("CSE", "ECE", "IT")
```

```
online <- c(52, 45, 30)
```

```
blended <- c(28, 35, 40)
```

```
face_to_face <- c(20, 20, 30)
```

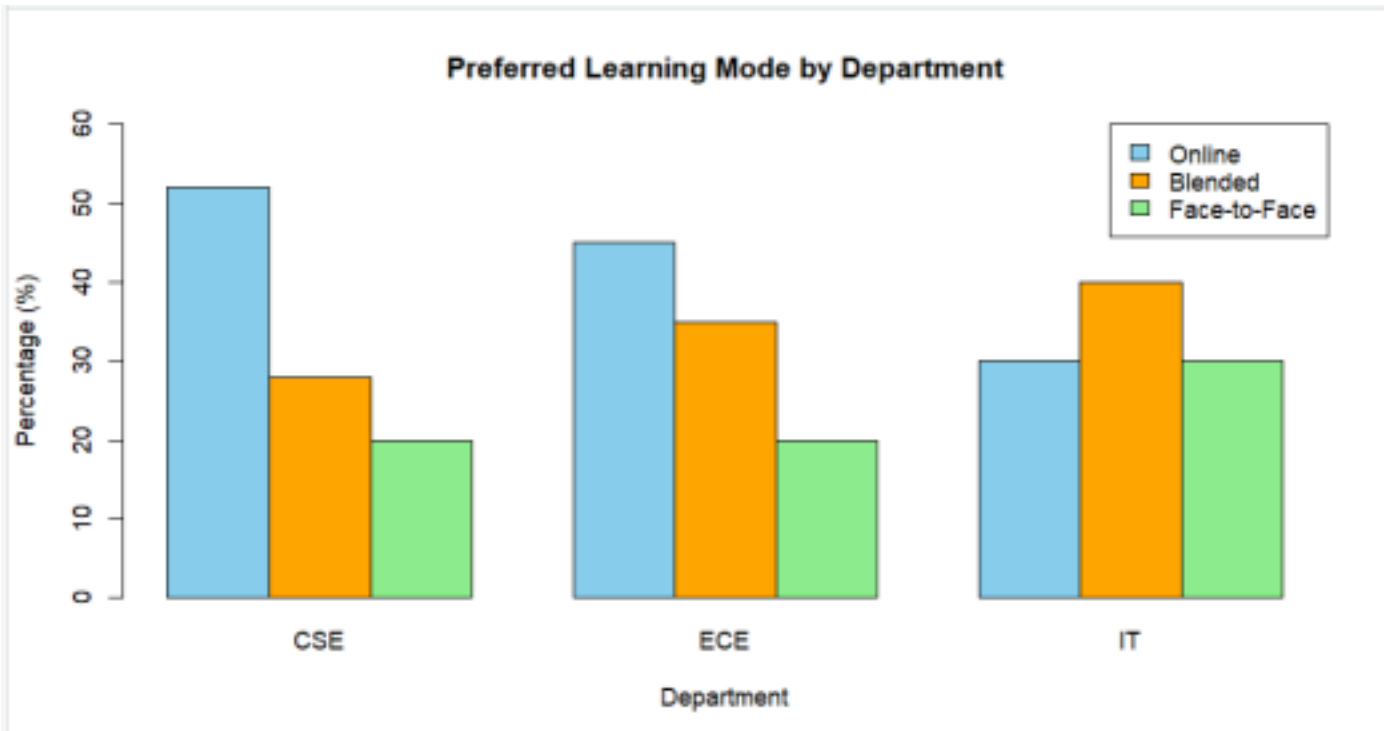
# Create matrix

```
data <- rbind(online, blended, face_to_face)
```

# Side-by-side bar chart

```
barplot(data,
  beside = TRUE,
  names.arg = dept,
  col = c("skyblue", "orange", "lightgreen"),
  ylim = c(0, 60),
  xlab = "Department",
  ylab = "Percentage (%)",
  main = "Preferred Learning Mode by Department")
```

```
legend("topright",
  legend = c("Online", "Blended", "Face-to-Face"),
  fill = c("skyblue", "orange", "lightgreen"))
```



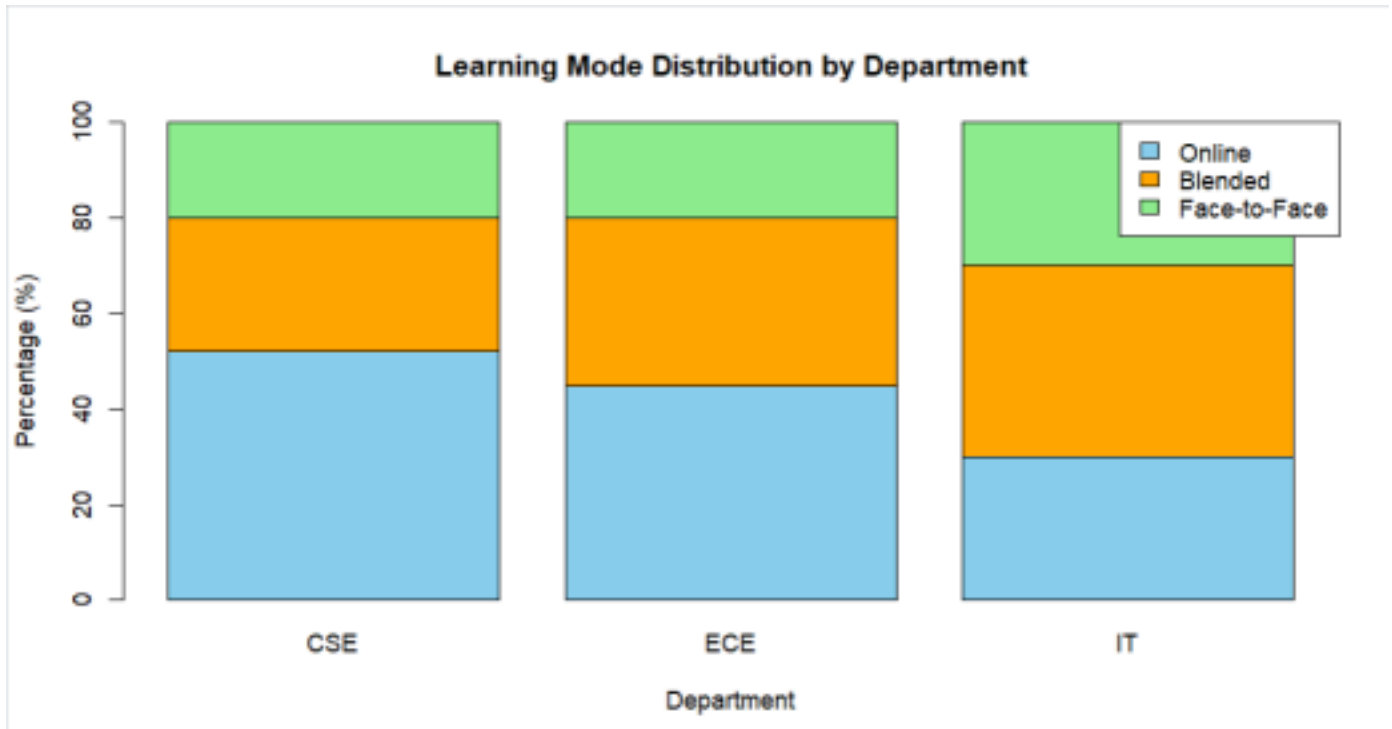
2. Explain which chart better supports comparison of individual proportions and why.

# 100% stacked bar chart

```
barplot(data,
  beside = FALSE,
  names.arg = dept,
  col = c("skyblue", "orange", "lightgreen"),
  ylim = c(0, 100),
  xlab = "Department",
  ylab = "Percentage (%)",
  main = "Learning Mode Distribution by Department")
```

```
legend("topright",
  legend = c("Online", "Blended", "Face-to-Face"),
  fill = c("skyblue", "orange", "lightgreen"))
```





### Which chart is better for comparing individual proportions?

The **side-by-side bar chart** is better for comparing individual proportions because all bars start from the same baseline of zero. This makes differences such as **CSE Online = 52%** and **ECE Online = 45%** easier to see accurately.

The 100% stacked bar chart is better for understanding the **overall composition** of each department, but comparing segments that are not aligned to the same baseline is more difficult.

3. Explain why separate pie charts may be less effective for comparing departments.

### Why separate pie charts are less effective

Separate pie charts make it difficult to compare similar-sized slices across departments because the slices have different angles and positions. Side-by-side bars provide a common baseline and make comparisons easier.

4. Identify one design choice that could make small proportional differences appear larger.

### Design choice that can exaggerate small differences

Using a **truncated y-axis** can make small differences appear much larger than they really are.

For example, displaying the y-axis from **40% to 55% instead of 0% to 100%** could make 45% and 52% look dramatically different.

**Conclusion:** Use the **side-by-side bar chart** when accurate comparison of individual percentages is the main goal, and the **100% stacked bar chart** when comparing the composition of each department.

**Key competency:** Proportions, side-by-side bars, stacked bars, pie charts, misleading proportions

## Question 4 – Nested Proportions using Treemap and Sunburst

**Scenario:** An e-commerce company analyzes annual revenue of ₹1.2 crore across product categories and subcategories. The dashboard must show both category contribution and internal composition.

| Category    | Subcategory | Revenue (₹ lakh) |
|-------------|-------------|------------------|
| Electronics | Mobiles     | 24               |
| Electronics | Laptops     | 18               |
| Electronics | Accessories | 8                |
| Home        | Furniture   | 14               |
| Home        | Appliances  | 10               |
| Home        | Decor       | 6                |
| Fashion     | Men         | 7                |
| Fashion     | Women       | 8                |
| Fashion     | Kids        | 5                |

### Tasks:

1. Construct a treemap showing category and subcategory revenue.

# Create data

```
data <- data.frame(
```

```
  Category = c(
```

```
    "Electronics", "Electronics", "Electronics",
```

```
    "Home", "Home", "Home",
```

```
    "Fashion", "Fashion", "Fashion"
```

```
),
```

```
  Subcategory = c(
```

```
    "Mobiles", "Laptops", "Accessories",
```

```
    "Furniture", "Appliances", "Decor",
```

```
    "Men", "Women", "Kids"
```

```
),
```

```
  Revenue = c(24, 18, 8, 14, 10, 6, 7, 8, 5)
```

)

# Treemap

treemap(

data,

index = c("Category", "Subcategory"),

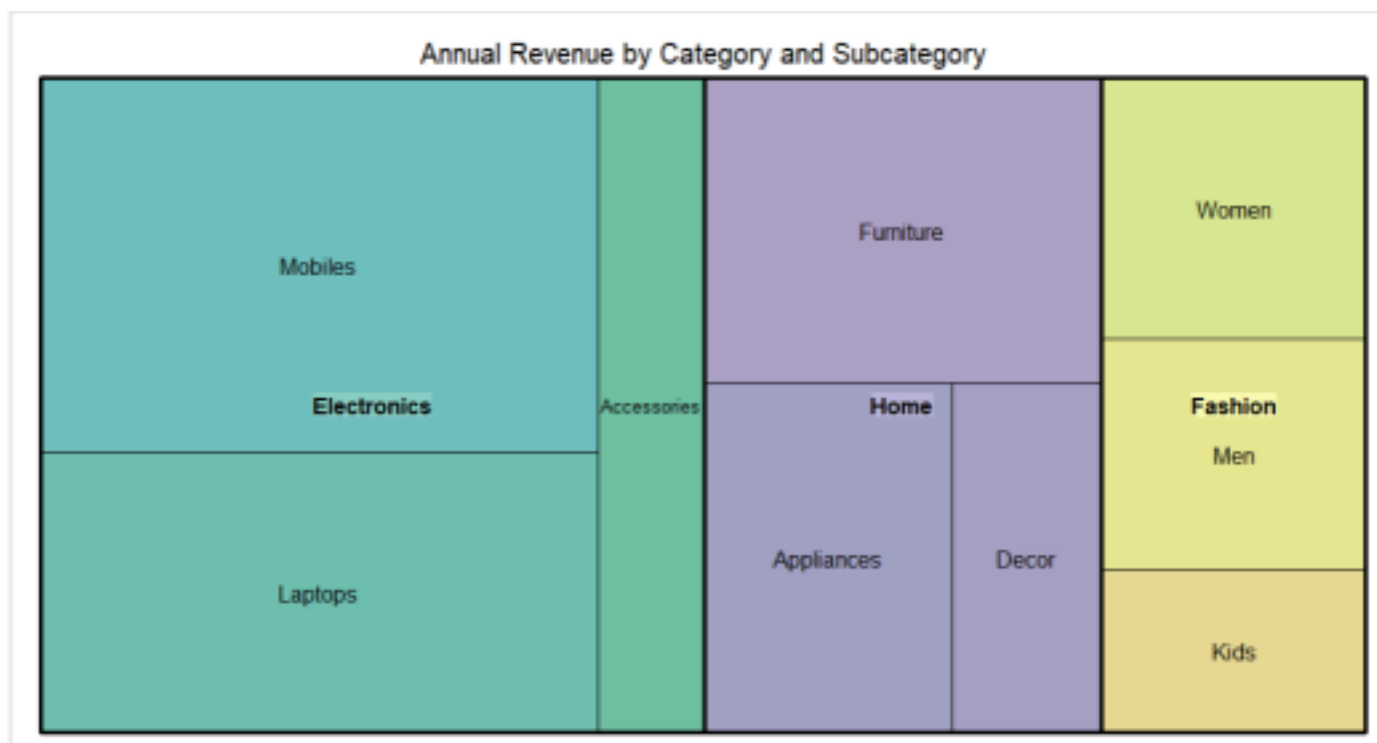
vSize = "Revenue",

type = "index",

title = "Annual Revenue by Category and Subcategory",

palette = "Set3"

)



2. Construct a sunburst chart representing the same hierarchy.

# Sunburst chart

fig <- plot\_ly(

data = data,

ids = ~Subcategory,

labels = ~Subcategory,

parents = ~Category,

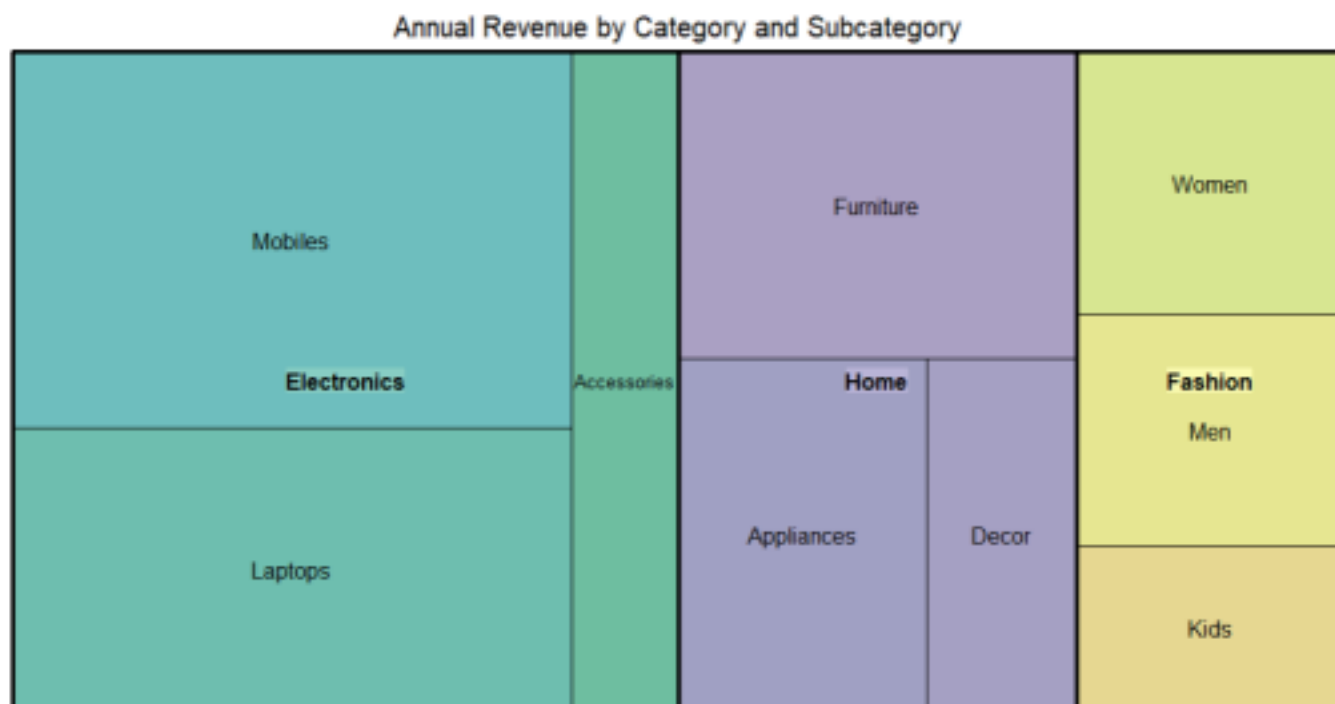
values = ~Revenue,

type = "sunburst"

)

```
fig <- fig %>%
  layout(
    title = "Revenue Distribution by Category and Subcategory"
  )
```

fig



3. Determine the largest category and calculate its percentage of total revenue.

```
category_total <- aggregate(
  Revenue ~ Category,
  data = data,
  sum
)
```

```
print(category_total)
```

```
Category Revenue
1 Electronics    50
2   Fashion     20
3    Home       30
> |
```

The totals are:

| Category    | Revenue (₹ lakh) | Percentage |
|-------------|------------------|------------|
| Electronics | 50               | 41.67%     |

|      |    |        |
|------|----|--------|
| Home | 30 | 25.00% |
|------|----|--------|

|         |    |        |
|---------|----|--------|
| Fashion | 20 | 16.67% |
|---------|----|--------|

**Note:** The supplied subcategory data totals ₹100 lakh (₹1 crore), not ₹120 lakh (₹1.2 crore). So, based on the actual table, Electronics contributes:

$$10050 \times 100 = 50\%$$

Therefore, **Electronics is the largest category and represents 50% of the provided revenue data.**

4. Compare treemap and sunburst for identifying nested proportions.

## Treemap vs Sunburst

- **Treemap:** Uses rectangular **area** to represent revenue. It is useful for quickly identifying the largest categories and subcategories.
- **Sunburst:** Uses concentric rings to show the **hierarchy**. The inner ring represents categories and the outer ring represents subcategories.
- **Treemap is better** for comparing sizes.
- **Sunburst is better** for understanding the parent-child relationship.

5. State one situation in which area encoding could mislead interpretation.

**Key competency:** Treemaps, sunburst charts, nested proportions, area encoding

## When area encoding can mislead

Area encoding can be misleading when two regions have similar areas but different shapes. Humans generally compare **length more accurately than area**, so small differences in rectangular areas may be difficult to judge precisely.

**Conclusion:** Use a **treemap for quick size comparison** and a **sunburst for hierarchical understanding**.

## Question 5 – Flow Proportions using Sankey Diagram / Parallel Sets

**Scenario:** A smart-city mobility study follows 200 commuters from residential zone through primary transport mode to destination zone. The planning team wants dominant flows and possible bottlenecks.

| Origin      | Mode  | Destination | Commuters |
|-------------|-------|-------------|-----------|
| Residential | Bus   | Central     | 35        |
| Residential | Bus   | Industrial  | 15        |
| Residential | Train | Central     | 25        |
| Residential | Train | Industrial  | 10        |
| Suburban    | Bus   | Central     | 20        |

|          |     |            |    |
|----------|-----|------------|----|
| Suburban | Bus | Industrial | 10 |
|----------|-----|------------|----|

|          |       |            |    |
|----------|-------|------------|----|
| Suburban | Train | Central    | 30 |
| Suburban | Train | Industrial | 15 |
| Rural    | Bus   | Central    | 12 |

#### DSA06 – Data Handling and Visualization | Assessment Tool: Experiment

|       |       |            |    |
|-------|-------|------------|----|
| Rural | Bus   | Industrial | 8  |
| Rural | Train | Central    | 10 |
| Rural | Train | Industrial | 10 |

#### Tasks:

1. Construct a Sankey diagram showing Origin → Mode → Destination.

Sankey Diagram: Origin → Mode → Destination

# Install once

```
install.packages("networkD3")
```

# Load package

```
library(networkD3)
```

# Create nodes

```
nodes <- data.frame(
```

```
  name = c(
```

```
    "Residential", "Suburban", "Rural",
```

```
    "Bus", "Train",
```

```
    "Central", "Industrial"
```

```
  )
```

```
)
```

```
# Create links
```

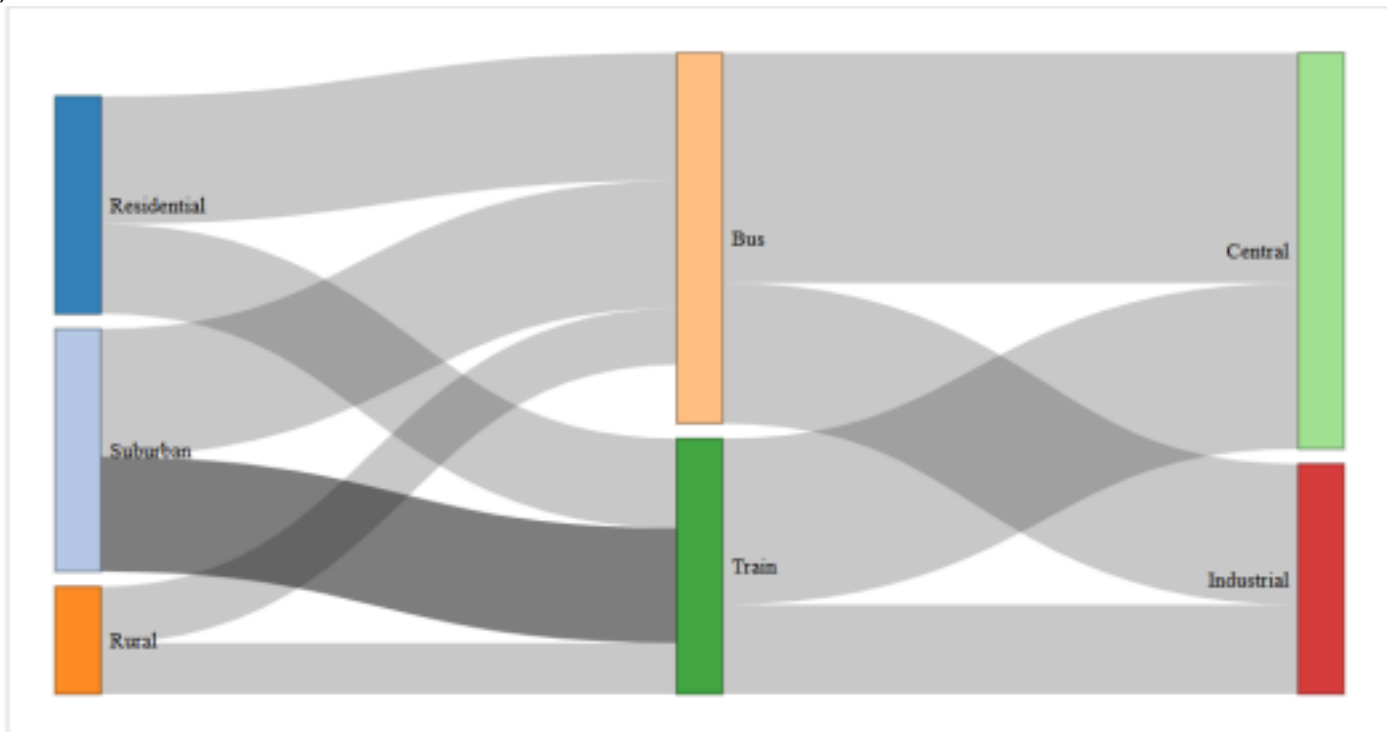
```
links <- data.frame(  
  source = c(  
  
    0, 0, 1, 1, 2, 2, 2,  
  
    3, 3, 4, 4, 3, 3, 4, 4  
  
  ),  
  
  target = c(  
  
    3, 3, 3, 3, 3, 3, 4,  
  
    5, 6, 5, 6, 5, 6, 5, 6  
  
  ),  
  
  value = c(  
  
    50, 15, 30, 10, 20, 12, 10,  
  
    35, 15, 25, 10, 20, 10, 30,  
  
    15 )  
  
)
```

```
# Create Sankey
```

```
diagram
```

```
sankeyNetwork(  
  
  Links = links,  
  
  Nodes = nodes,  
  
  Source = "source",  
  
  Target = "target",  
  
  Value = "value",  
  
  NodeID = "name",  
  
  fontSize = 12,  
  
  nodeWidth = 30
```

)



2. Construct a parallel-sets representation of the same categorical flows.

```
data <- data.frame(
```

```
  Origin = c(
```

```
    "Residential", "Residential", "Residential", "Residential",
    "Suburban", "Suburban", "Suburban", "Suburban",
    "Rural", "Rural", "Rural", "Rural"
```

```
),
```

```
  Mode = c(
```

```
    "Bus", "Bus", "Train", "Train",
    "Bus", "Bus", "Train", "Train",
    "Bus", "Bus", "Train", "Train"
```

```
),
```

```
  Destination = c(
```

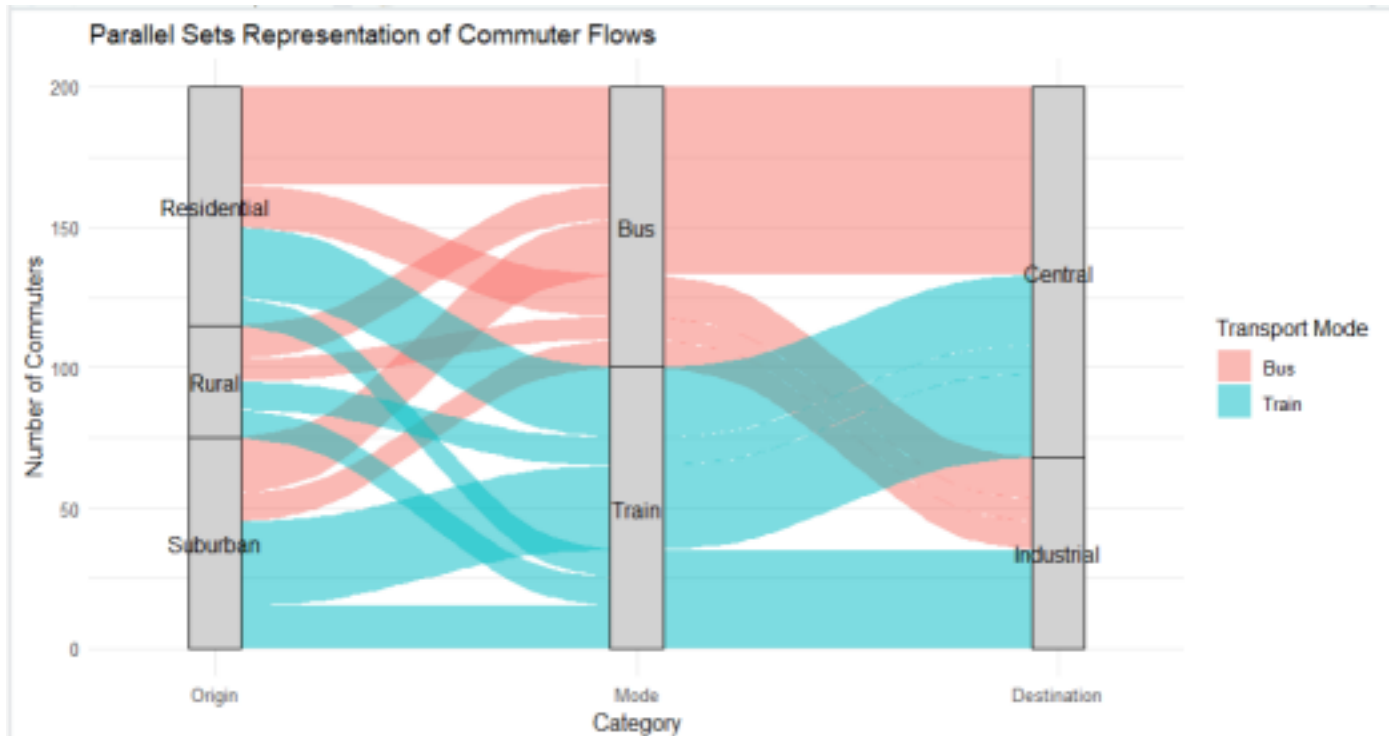
```
    "Central", "Industrial", "Central", "Industrial",
    "Central", "Industrial", "Central", "Industrial",
    "Central", "Industrial", "Central", "Industrial"
```

```
),
```

```
  Commuters = c(35, 15, 25, 10, 20, 10, 30, 15, 12, 8, 10, 10)
```

```
)
```





3. Identify the three largest flows and calculate their percentages of all commuters.

### Three Largest Flows

The largest individual flows are:

| Rank | Origin → Mode → Destination   | Commuters | Percentage |
|------|-------------------------------|-----------|------------|
| 1    | Residential → Bus → Central   | 35        | 17.5%      |
| 2    | Suburban → Train → Central    | 30        | 15%        |
| 3    | Residential → Train → Central | 25        | 12.5%      |

Percentage is calculated as:

$$\text{Percentage} = \frac{\text{Number of Commuters}}{\text{Total Number of Commuters}} \times 100$$

Together, these three flows account for:

$$\frac{35 + 30 + 25}{200} \times 100 = 45\%$$

4. Compare Sankey and parallel sets for flow magnitude and category relationships.

### Sankey vs Parallel Sets

**Sankey diagram:**

- Best for showing **flow magnitude**.
- Link width represents the number of commuters.
- Makes dominant routes and bottlenecks easy to identify.

#### Parallel sets:

- Best for showing **relationships between categorical variables**.
- Makes it easier to see how Origin, Mode and Destination categories interact.
- Useful when there are many categorical dimensions.

5. Explain one way poor link widths or category ordering could mislead the viewer.

**Key competency:** Sankey diagrams, parallel sets, flow proportions, categorical relationships

### How visualization can mislead

If the **link widths are not proportional to the actual commuter counts**, a flow may appear more or less important than it really is.

Similarly, poor **category ordering** can create unnecessary crossings, making some relationships appear stronger or more complicated than they actually are.

### Conclusion

The **Residential** → **Bus** → **Central** flow is the largest individual flow with **35 commuters (17.5%)**. A Sankey diagram is preferable for understanding **flow magnitude**, while parallel sets are useful for examining **categorical relationships**.

### Suggested Evaluation Rubric – Experiment

| Criterion            | Excellent (4) Good (3)                          | Satisfactory (2)      | Needs Improvement (1) |
|----------------------|-------------------------------------------------|-----------------------|-----------------------|
| Data preparation     | Correct and complete Minor issue                | Several issues        | Incorrect/incomplete  |
| Visualization choice | Highly appropriate<br>Appropriate and justified | Partially appropriate | Poor choice           |
| Technical execution  | Accurate and clear Mostly accurate              | Some errors           | Major errors          |
| Interpretation       | Insightful and<br>evidence-based Correct        | Basic                 | Incorrect/unsupported |

Misleading-design analysis

Bias clearly identified and explained  
Bias identified Limited explanation

**Suggested marks:** 20 marks per question; 100 marks total. Level: Competitive / advanced undergraduate. DSA06 – Data Handling and Visualization | Assessment Tool: Experiment